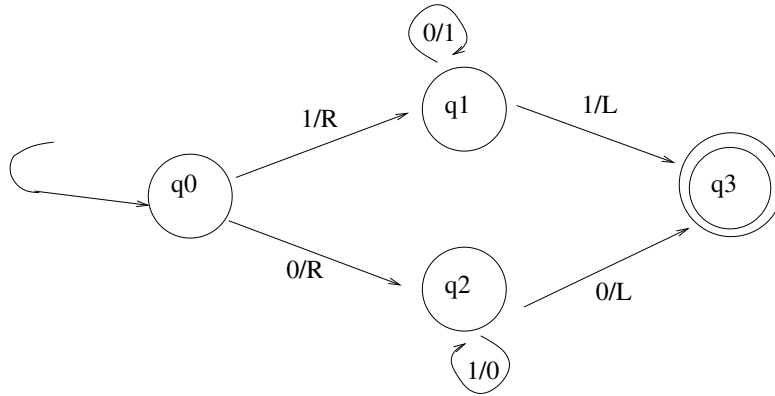


COMPX361-22B Logic and Computation

Assignment 3

Due 4PM, Friday 30 September (end of Week 9)

1. [4+2 marks.] Consider the Turing machine M drawn below.



- (a) Using our table from class (given below), give the first 60 bits in the binary encoding of M . (First normalize M in the obvious way, so that q_i becomes state i .)
- (b) Briefly describe what M does: that is, when does it complete a computation and what does it do overall when it does? Just describe the output for a given input, no need to provide a detailed commentary.

symbol	code
0	10000
1	10001
2	10010
$\vdots$	$\vdots$
9	11001
b	11010
$\perp$	11011
L	11100
R	11101
,	11110
/	11111

2. [2+2+2 marks.] For each of the following grammars G , by a combination of experimentation and examination of the production rules, determine the associated language $L(G)$. (A description in words may be necessary in some cases.) There is no need to write your reasoning.

(a) $G = (V, T, S, P)$, where $V = \{S, A\}$, $T = \{a, b\}$ and

$$P = \{S \Rightarrow aA|b, A \Rightarrow aa\}.$$

(b) $G = (V, T, S, P)$, where $V = \{S, A, B\}$, $T = \{a, b\}$ and

$$P = \{S \Rightarrow AB|AA, A \Rightarrow aB|ab, B \Rightarrow b\}.$$

(c) $G = (V, T, S, P)$, where $V = \{S, X, Y\}$, $T = \{x, y, z, =\}$ and

$$P = \{S \Rightarrow XYX, X \Rightarrow XX|x|y|z, Y \Rightarrow =\}.$$

3. [6 marks.] Consider the following regular grammar:

$$G = (V, T, S, P), \text{ where } V = \{S, A, B\}, T = \{0, 1\},$$

and

$$P = \{S \Rightarrow 0A|\lambda, A \Rightarrow 0A|1B, B \Rightarrow 0|0A\}.$$

Using the method given in the lectures, find a DFSA that accepts the regular language it generates. (This means first converting the grammar into an NFSA and then using the subset method to find a DFSA.)

4. [3 marks.] Let $G = (V, T, S, P)$ be the context-free grammar in which $T = \{a, b, c\}$, $V = \{S, A, B\}$ and

$$P = \{S \Rightarrow A, A \Rightarrow AA|aAc|B, B \Rightarrow b\}.$$

Give a parse tree and a derivation for the string $abcabc$.

5. [2 marks.] Let $G = (V, T, S, P)$ be the context-sensitive grammar in which $T = \{0, 1\}$, $V = \{S, A\}$ and

$$P = \{S \Rightarrow 0A1|1A0|\lambda, 0A1 \Rightarrow 01A01|01, 1A0 \Rightarrow 10A10|10\}.$$

Give a derivation for the string 010101.

(turn over)

6. [4+3 marks.] For the well-formed formulae below, give as much of the truth table as is needed to determine its logical status (whether it is valid, and whether it is satisfiable or unsatisfiable). Justify your answer (in terms of the existence of models and/or counterexamples).

(a) $((p \rightarrow q) \rightarrow r) \leftrightarrow (p \rightarrow (q \rightarrow r))$

(b) $(p \leftrightarrow q) \rightarrow (p \oplus q)$